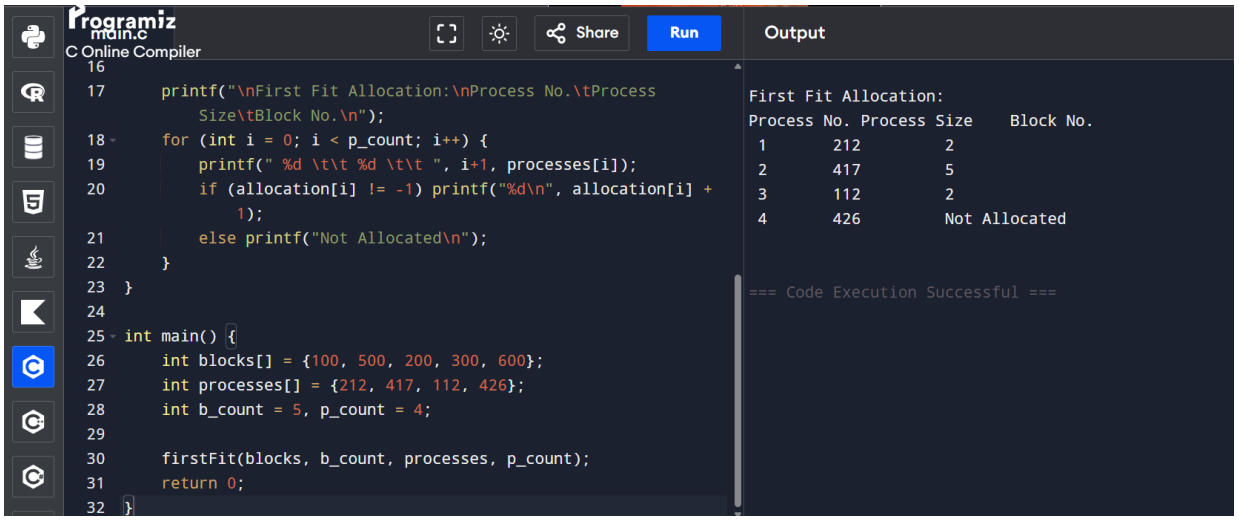
EXPERIMENT 13:

13. Construct a C program for implementation the various memory allocation strategies. .



PSEUDO CODE :

1. Input block sizes and process sizes.
2. First Fit: Allocate process to the FIRST block large enough.
3. Best Fit: Allocate process to the SMALLEST block large enough.
4. Worst Fit: Allocate process to the LARGEST block available.
5. Print allocation result for each process.

CODE:

```
#include <stdio.h>

void firstFit(int blocks[], int b_count, int processes[], int p_count) {
    int allocation[10];
    for(int i = 0; i < p_count; i++) allocation[i] = -1;

    for (int i = 0; i < p_count; i++) {
        for (int j = 0; j < b_count; j++) {
            if (blocks[j] >= processes[i]) {
                allocation[i] = j;
                blocks[j] -= processes[i];
                break;
            }
        }
    }

    printf("\nFirst Fit Allocation:\nProcess No.\tProcess Size\tBlock No.\n");
    for (int i = 0; i < p_count; i++) {
        printf(" %d \t\t %d \t\t ", i+1, processes[i]);
        if (allocation[i] != -1) printf("%d\n", allocation[i] + 1);
        else printf("Not Allocated\n");
    }
}

int main() {
    int blocks[] = {100, 500, 200, 300, 600};
    int processes[] = {212, 417, 112, 426};
```

```
    int b_count = 5, p_count = 4;


    firstFit(blocks, b_count, processes, p_count);

    return 0;

}
```